

KRASNOYARSK MININO, Russian Federation

WMO: 295700

Lat: **56.067N**Lon: **92.733E**Elev: **770**StdP: **14.29**Time Zone: **7.00 (E07)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-28.8	-38.1	0.6	-33.7	-33.1	0.9	-28.7	19.0	14.3	17.6	16.2	0.9	230	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.8	66.6	81.5	65.4	78.3	63.8	69.3	80.5	67.4	77.8	65.7	75.1	5.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	96.6	74.3	63.7	90.8	72.3	61.9	85.2	70.9	34.0	80.2	32.4	78.0	31.0	75.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.0	13.8	11.9	DB	-34.4	91.0	7.6	3.2	-39.9	93.3	-44.3	95.2	-48.6	97.0	-54.1	99.3	
			WB	-34.7	71.9	7.4	2.2	-40.0	73.5	-44.3	74.8	-48.4	76.0	-53.8	77.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	1.6	8.6	23.6	38.2	50.0	63.1	66.1	61.0	48.8	35.6	17.6	7.4	
	DBStd	24.98	17.58	14.96	11.60	9.61	9.92	7.17	5.53	6.67	8.11	10.18	14.99	15.88	
	HDD50	6847	1499	1160	819	367	128	5	0	5	118	454	972	1320	
	HDD65	11039	1964	1580	1284	803	475	117	53	158	487	911	1422	1785	
	CDD50	1475	0	0	0	14	127	397	499	347	83	8	0	0	
	CDD65	193	0	0	0	0	9	59	88	35	2	0	0	0	
	CDH74	1801	0	0	0	2	111	655	712	294	27	0	0	0	
	CDH80	441	0	0	0	0	24	167	176	71	3	0	0	0	
Wind	WSAvg	4.9	4.8	4.9	5.5	5.9	5.5	4.4	3.8	3.9	4.6	5.4	5.3	5.2	
Precipitation	PrecAvg	17.5	0.6	0.6	0.7	0.9	1.6	2.4	2.4	2.7	1.9	1.4	1.2	1.1	
	PrecMax	21.8	1.3	1.2	1.5	1.8	3.4	4.6	4.6	4.8	3.8	2.6	2.5	2.2	
	PrecMin	13.2	0.0	0.1	0.0	0.1	0.3	0.9	0.1	0.7	0.4	0.4	0.4	0.0	
	PrecStd	2.2	0.3	0.3	0.4	0.5	0.9	0.9	1.1	1.0	0.8	0.6	0.5	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	39.3	52.2	70.1	83.5	88.9	88.7	86.8	78.0	63.7	51.0	37.4	
		MCWB	31.1	32.3	40.8	51.0	60.1	66.6	69.7	69.2	61.5	50.1	42.9	31.1	
	2%	DB	31.0	35.1	46.4	63.4	76.7	84.6	84.7	81.2	70.9	58.5	44.6	33.1	
		MCWB	27.3	29.2	37.2	47.8	57.0	65.8	67.5	66.7	58.1	48.4	38.2	29.2	
	5%	DB	27.8	32.0	42.6	58.5	72.1	81.1	81.1	76.8	66.2	54.0	40.2	30.3	
		MCWB	24.2	27.7	35.2	46.0	55.5	64.1	66.7	64.7	55.9	46.2	34.9	27.2	
	10%	DB	24.6	28.5	38.8	54.2	67.6	77.6	77.8	73.1	61.7	49.9	36.2	27.3	
		MCWB	21.5	25.0	32.7	43.6	53.1	62.6	65.1	62.5	53.7	43.5	32.6	24.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.4	33.3	41.7	52.5	62.6	70.2	72.7	71.0	62.8	52.7	43.7	32.2
			MCDB	34.2	39.0	50.3	66.2	79.1	83.5	83.3	82.4	73.8	60.4	49.6	35.0
2%		WB	27.7	30.1	38.3	49.5	59.3	68.0	69.9	67.9	59.2	49.5	38.6	29.5	
		MCDB	30.9	34.1	45.2	61.2	73.1	80.6	80.6	78.1	69.2	56.8	44.1	32.9	
5%		WB	24.2	27.6	35.6	46.6	56.6	66.1	68.0	65.6	56.8	46.6	35.1	27.3	
		MCDB	27.4	31.6	41.8	57.4	69.2	77.2	78.0	74.7	64.2	53.2	39.4	30.3	
10%		WB	21.5	24.8	33.1	43.7	53.9	64.1	66.4	63.7	54.4	43.9	32.6	24.5	
		MCDB	24.6	28.6	38.5	53.7	66.1	74.6	75.5	71.9	61.0	49.6	36.1	27.3	
Mean Daily Temperature Range	5% DB	MDBR	12.7	14.7	16.7	17.4	19.1	19.9	17.8	17.2	16.1	13.0	12.0	12.4	
		MCDBR	12.0	15.1	20.2	25.8	26.4	27.4	23.9	24.8	24.7	19.5	12.1	12.7	
		MCWBR	10.4	11.9	13.6	14.1	13.3	11.9	10.2	11.7	13.0	11.9	8.2	10.8	
	5% WB	MCDBR	12.1	14.1	18.8	23.8	23.8	22.6	20.9	21.7	22.3	18.3	11.8	13.2	
		MCWBR	10.8	11.5	13.2	13.8	12.6	10.9	10.1	11.0	12.4	11.6	8.5	11.7	
Clear-Sky Solar Irradiance	taub	0.284	0.317	0.364	0.396	0.414	0.410	0.433	0.414	0.371	0.352	0.319	0.285		
	taud	2.025	2.006	1.966	2.128	2.192	2.300	2.249	2.270	2.339	2.237	2.065	1.971		
	Ebn at Noon	183	225	244	258	262	265	255	252	246	215	165	144		
	Edh at Noon	21	32	43	42	42	38	40	36	30	25	20	17		
All-Sky Solar Radiation	RadAvg	252	566	994	1363	1563	1884	1680	1346	864	458	228	149		
	RadStd	28	36	75	89	166	157	113	118	114	36	31	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD6